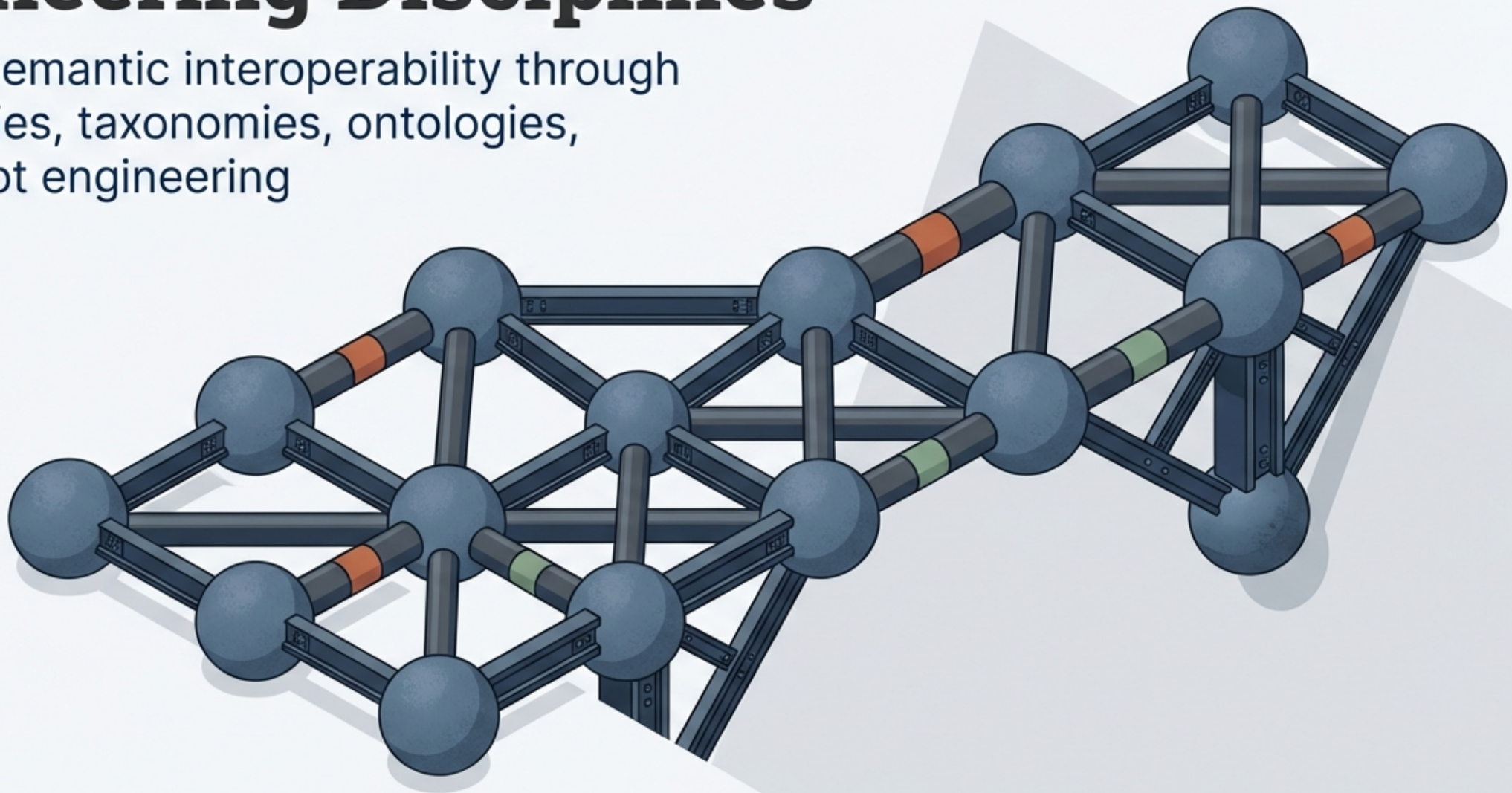
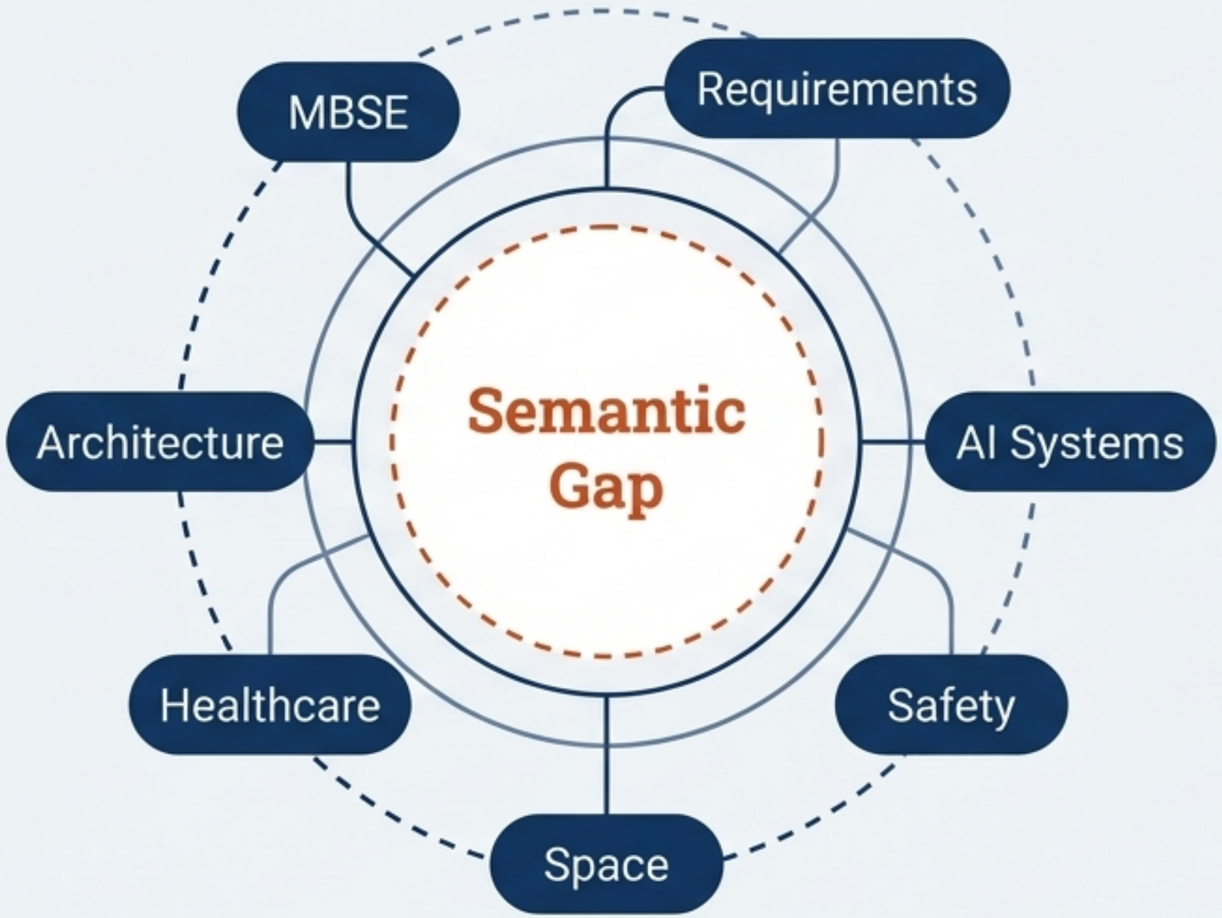
Bridging Knowledge & AI Across Engineering Disciplines

Enabling semantic interoperability through vocabularies, taxonomies, ontologies, and prompt engineering



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50+ Working Groups. One Language Barrier.



Incompatible Terminologies

'Requirement' means entirely different things in the Requirements WG, MBSE Initiative, and Healthcare WG.

Redundant Development

Multiple WGs independently build overlapping frameworks without coordination.

AI Hallucinations

Without semantic grounding, AI systems fail to accurately leverage Systems Engineering knowledge.

Missed Insights

No infrastructure exists to synthesize expertise across disciplines.

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What Is At Stake for Enterprise Knowledge?



Lost Innovation

Critical cross-domain insights require multiple WGs, but no infrastructure enables this collaboration.



AI Readiness Gap

Organizations leveraging AI-ready knowledge structures are outpacing those with fragmented data.



Diminished Impact

Siloed working groups deliver significantly less value than a connected ecosystem.

The question isn’t whether to act—it’s how quickly we can enable interoperability. Every month of delay compounds fragmentation.

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The Shift from Findability to Semantic Discovery



Findability

Researchers seek to find exactly what they are looking for.

(Simple keyword matching)



Semantic Discovery

Finding valuable connections you weren't actively looking for.

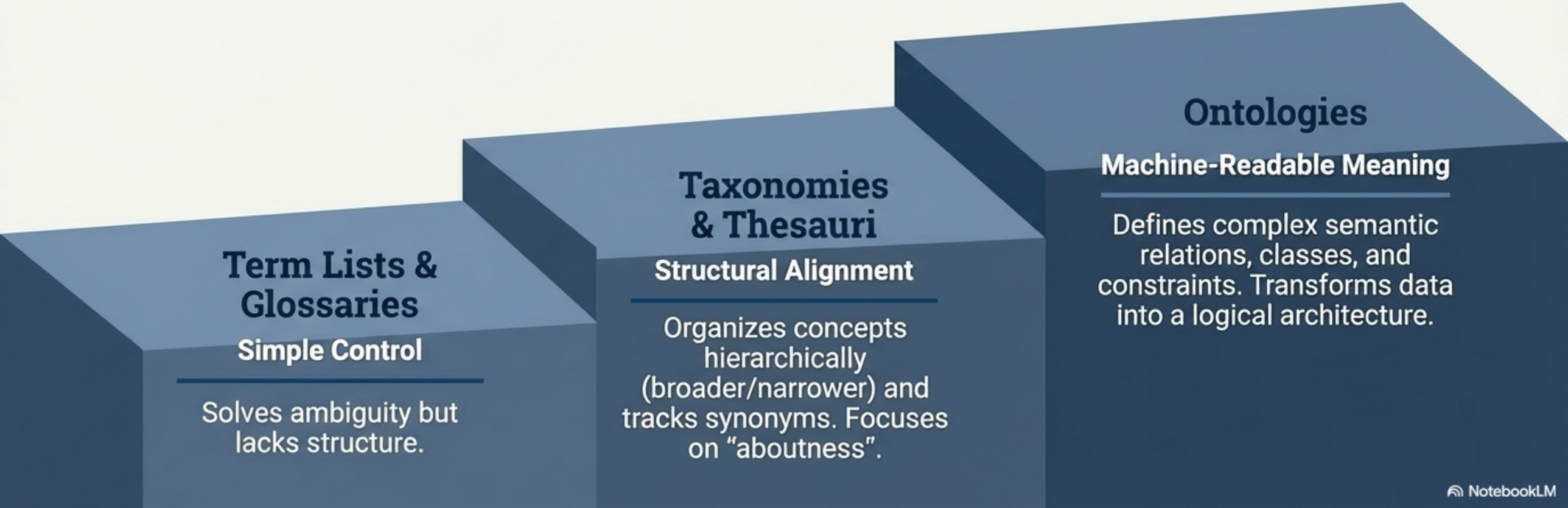
(Learning broader connections, identifying cross-domain correlations, and discovering related subject areas)

Takeaway: Semantic infrastructure moves organizations from simple **retrieval to proactive knowledge discovery**.

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The Knowledge Organization Spectrum

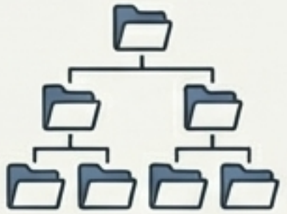
Core Principle: The focus must be on "Things, not strings"—managing explicit concepts rather than relying on ambiguous text labels.



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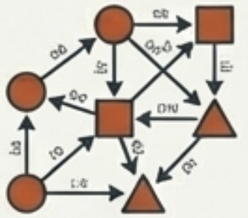
Delineating Taxonomies and Ontologies

Taxonomies



- **Hierarchy**: Generic/specific (Is-A) and whole/part parent-child relationships.
- **Labels**: Highly rich in synonym control (preferred labels, alternative labels, hidden labels).
- **Purpose**: Tagging, categorization, and content retrieval.

Ontologies



- **Relations**: Complex, semantic bi-directional relations (e.g., ‘allocates to’, ‘implements function’).
- **Structure**: Classes, sub-classes, individuals, and specific data attributes.
- **Purpose**: Machine-readable logic, inferencing, and semantic modeling.

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The Semantic Web Standards Stack

Top Layer: OWL (Web Ontology Language)

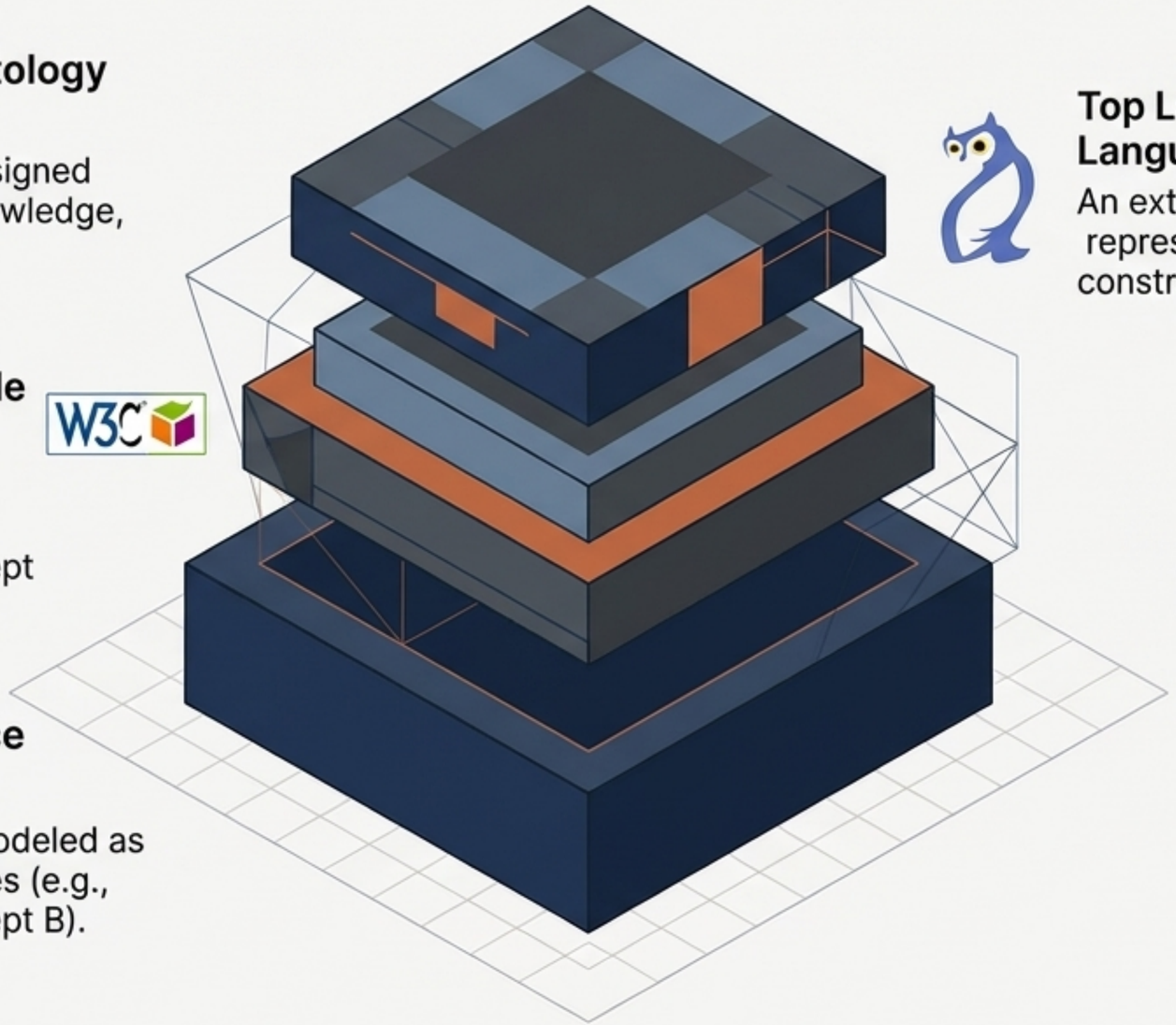
An extension of RDF/RDFS designed to represent rich, complex knowledge, constraints, and classes.

Middle Layer: SKOS (Simple Knowledge Organization System)

Built on RDF for publishing taxonomies and sharing concept schemes.

Base Layer: RDF (Resource Description Framework)

The bedrock. Information is modeled as subject-predicate-object triples (e.g., Concept A - relates to - Concept B).



Top Layer: OWL (Web Ontology Language)

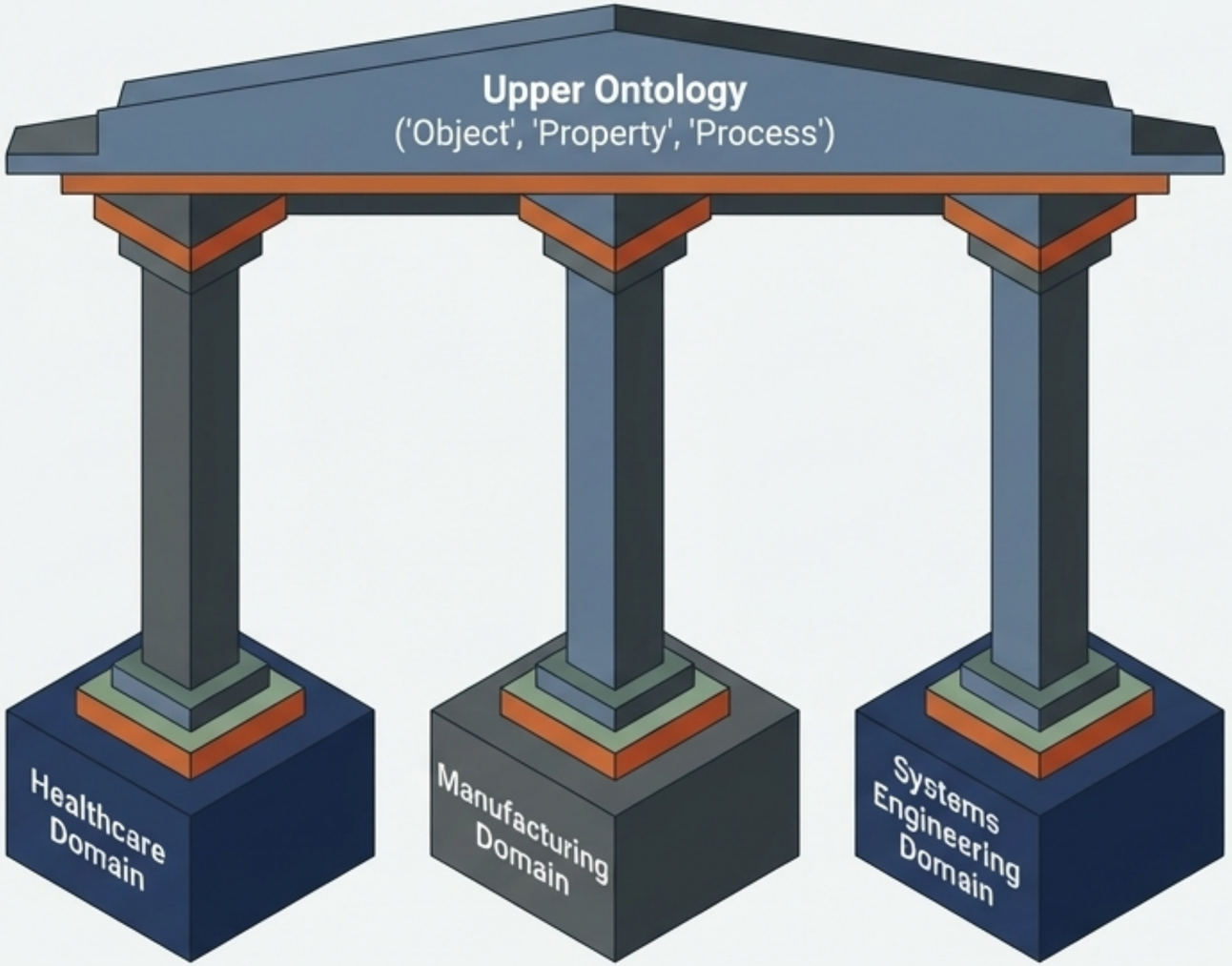
An extension of RDF/RDFS designed to represent rich, complex knowledge, constraints, and classes.

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Upper Ontologies: The Universal Translators

Role

Provides a common starting point and broad semantic interoperability across disparate domains.



BFO (Basic Formal Ontology)

Extensively used in defense (Common Core), manufacturing (IOF), and UN Sustainable Development Goals.

gist

A minimalist upper ontology designed specifically for enterprise information systems.

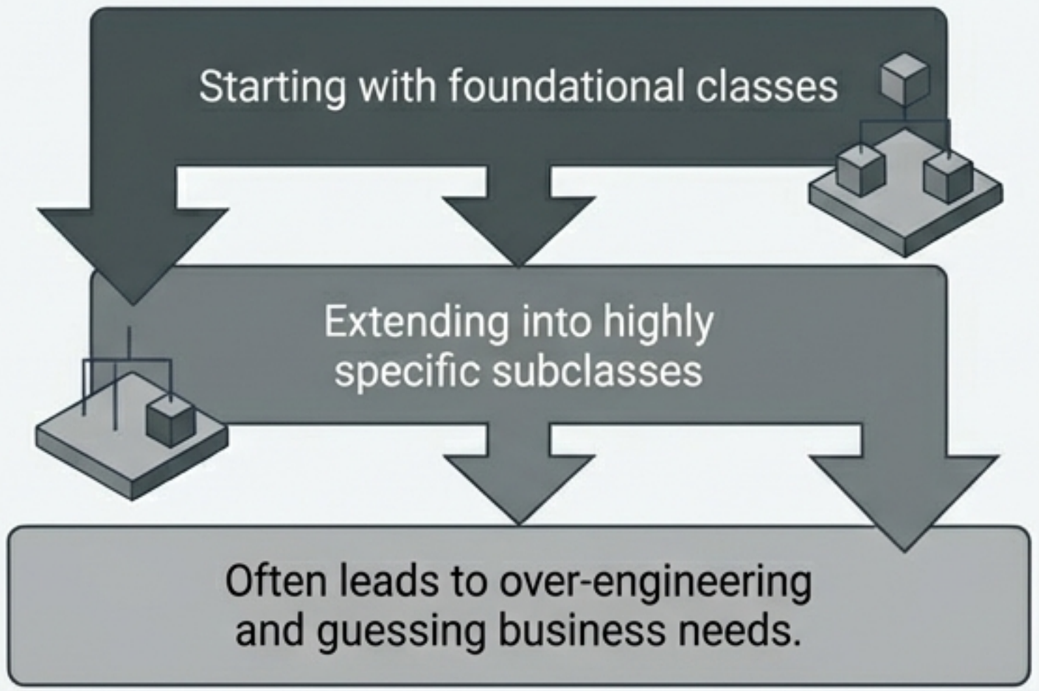
SUMO

The largest formal public ontology mapped to the WordNet lexicon.

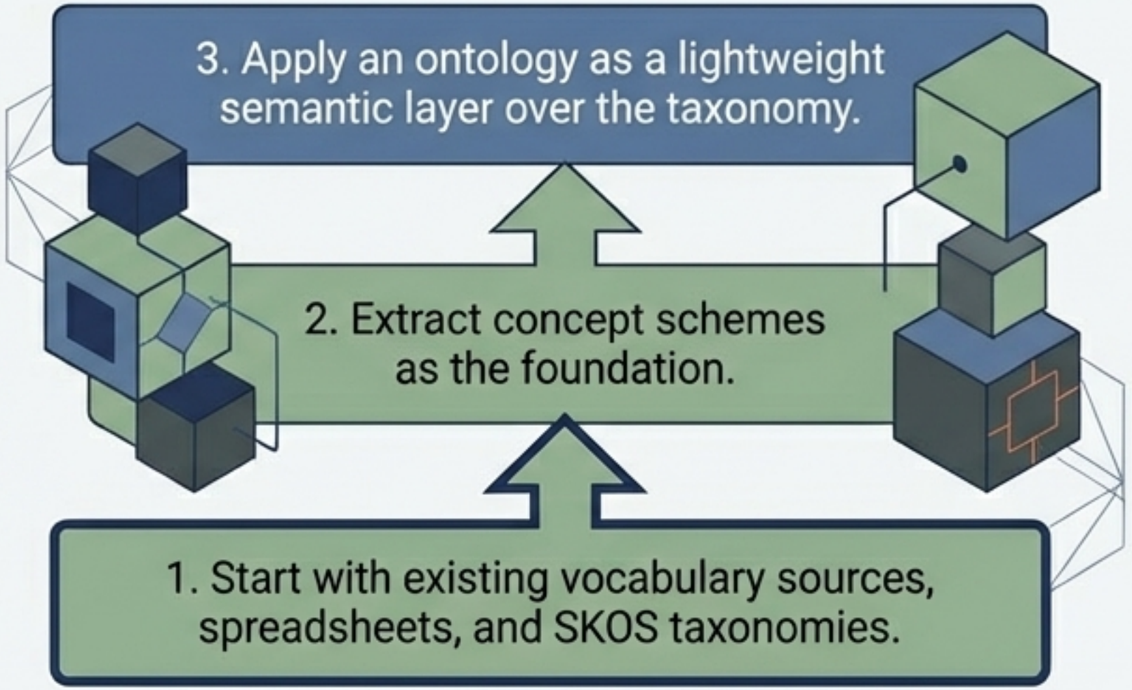
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Pragmatic Architecture: Taxonomy-Driven Ontology Design

Pathway 1: Top-Down Approach



Pathway 2: Bottom-Up (Taxonomy-Driven) Approach

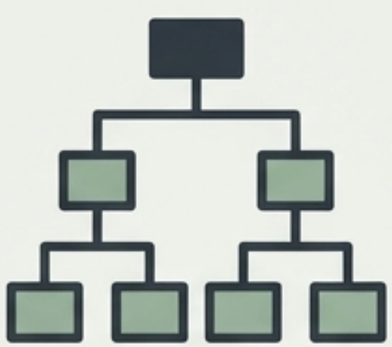


The Benefit: Drastically more flexible.

Taxonomies evolve frequently with language; ontologies remain stable for logical analysis.

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The Knowledge Graph Equation



Taxonomy (SKOS)

The controlled vocabulary and structural concepts.

+



Ontology (OWL)

The semantic layer, classes, and complex rules.

+



Instance Data

The specific facts and entities extracted from databases and documents.

=

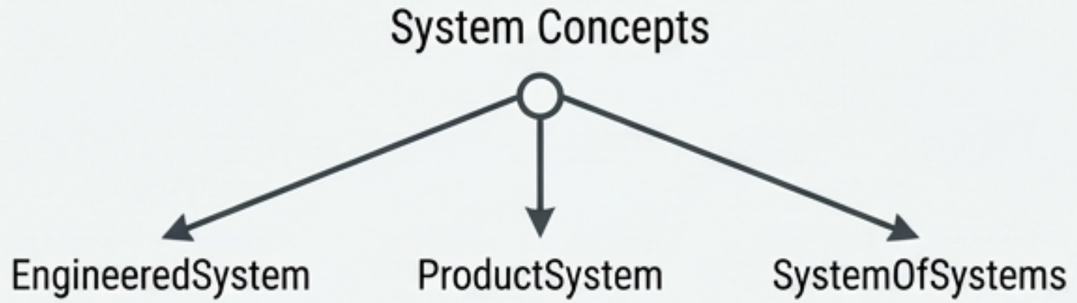
Enterprise Knowledge Graph

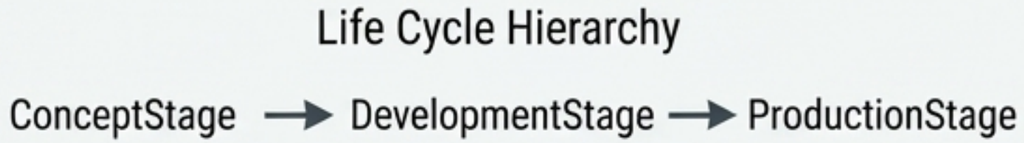
A Knowledge Graph unites your domain terminology, semantic logic, and raw data into a single, machine-readable fabric.

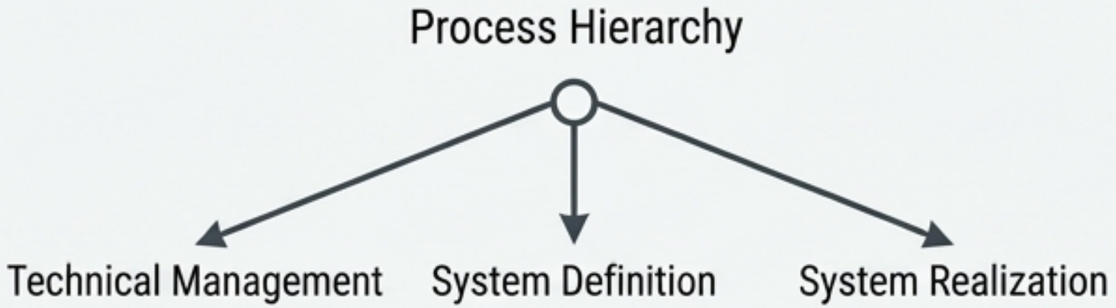
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Case Study: The INCOSE Core Ontology

Standard Checked: ISO/IEC/IEEE 15288:2023







Relationship Pattern Example

```
Customer1 expresses StakeholderReq1
SystemReq1 derivedFrom StakeholderReq1
SystemReq1 allocatedTo ProcessingSubsystem
```

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Proving Machine Readability with SPARQL

```
SELECT ?req ?arch ?element WHERE {
  ?req a incose:SystemRequirement ;
         incose:allocatedTo ?element .
  ?arch incose:implementsFunction ?element .
  :MySystem incose:hasArchitecture ?arch .
}
```



Plain English Translation:

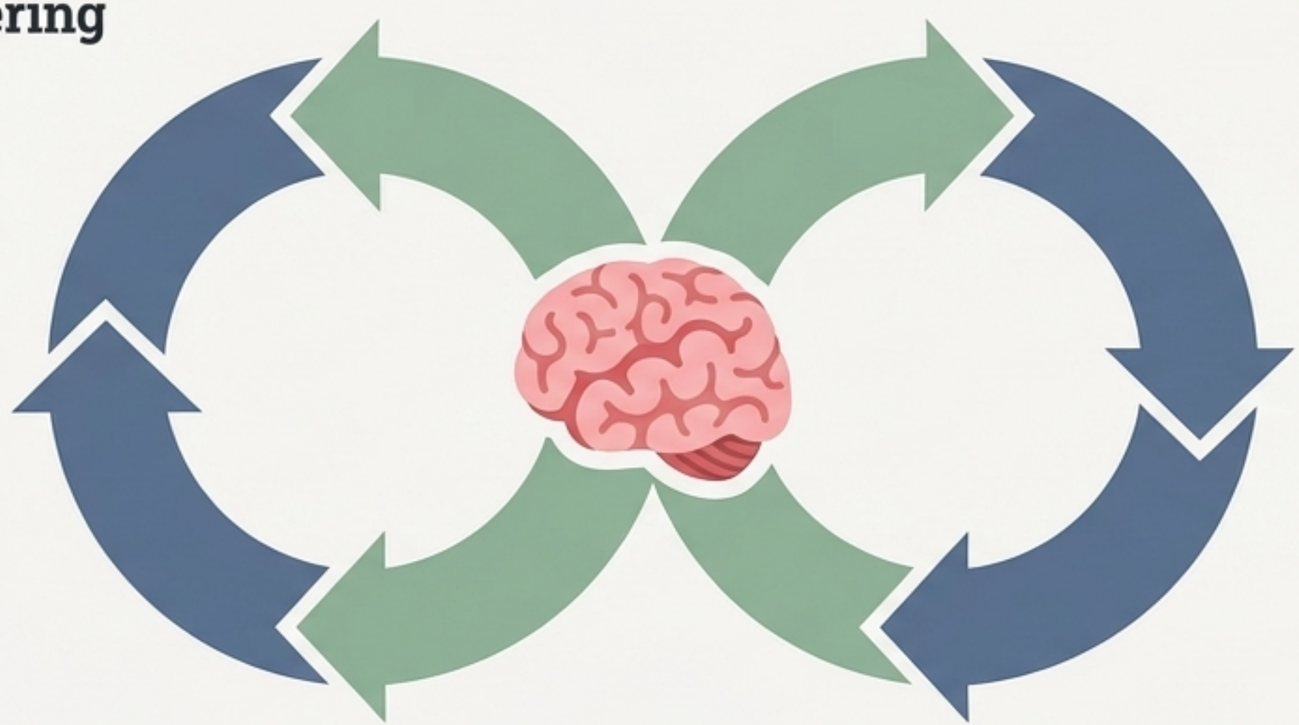
Trace all system requirements directly to the architecture elements that implement them across the entire product lifecycle.

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Knowledge Engineering + Prompt Engineering

Knowledge Engineering (The Grounding)

- Standardized vocabularies across all domains.
- Hierarchical classification (Taxonomies).
- Formal semantic relationships (Ontologies).



Prompt Engineering (The Execution)

- **RAG 2.0**: Generating AI responses enhanced directly by the Knowledge Graph.
- **Semantic Guardrails:** Forcing AI outputs to strictly align with established standards.
- **Human-AI Patterns:** Delivering explainable, trustworthy AI across a Unified Knowledge Fabric.

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The Enterprise ROI of Semantic Alignment



Understanding

Shared conceptual models bypass the need for constant translation.



Coordination

Enterprise knowledge graphs provide hard infrastructure for cross-group initiatives.



Communication

A unified semantic layer enables unambiguous communication.



Impact

Trustworthy, AI-enhanced decision-making.

50+
Working Groups Connected

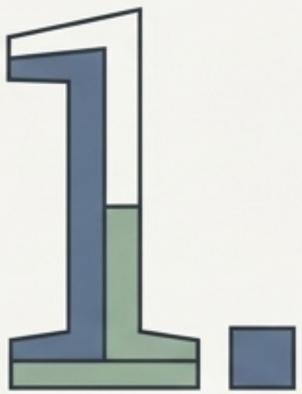
1
Unified Language

4.8x
First-Year ROI

40%
AI Accuracy Gain via RAG 2.0

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Join Us in Building the Unified Knowledge Fabric



Participate

Share your working group's key terms, core concepts, and data pain points.



Collaborate

Join cross-working group sessions to actively map connections and resolve semantic gaps.



Champion

Serve as the liaison between the Knowledge Systems Working Group and your specific discipline.

Let's talk after the session. | KSWG Team

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